

1. What is Linear Regression?

Linear Regression is a supervised machine learning algorithm used to model the relationship between one or more input variables (features) and a continuous output variable (label).

It assumes that this relationship can be approximated by a **linear function**.

Basic idea:

We try to draw the **best possible straight line** that fits the data and can be used to make predictions.

2. Supervised Learning Context

Linear Regression belongs to **supervised learning**, meaning:

- We have **input data (X)**
 - We have **correct answers (y)**
 - The model learns a mapping from $X \rightarrow y$
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3. Core Concepts and Terminology

◆ Feature

A **feature** is an input variable used to make predictions.

Examples:

- House size (m²)
- Years of experience
- Total bill amount

Mathematically:

Features are usually represented as **X**

◆ Label

A **label** (also called target) is the value we want to predict.

Examples:

- House price
- Salary
- Tip amount

Mathematically:

Labels are represented as **y**

◆ Prediction

A **prediction** is the value produced by the model based on the input features.

Mathematically:

$$\hat{y} = \text{model}(X)$$

The goal of learning is to make predictions **as close as possible** to the true labels.

4. Linear Regression Model Formula

Single feature:

$$\hat{y} = w \cdot x + b$$

Multiple features:

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

Where:

- **w (weights)** determine the importance of each feature
 - **b (bias)** shifts the prediction up or down
 - **$\hat{y}$** is the predicted value
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5. Weight (Coefficient)

A **weight** represents how strongly a feature influences the prediction.

Example:

- If $w = 3$, then increasing x by 1 increases prediction by 3

Weights are learned during training.

6. Bias (Intercept)

The **bias** is a constant added to the prediction.

Purpose:

- Allows the model to make predictions even when all feature values are zero
- Shifts the regression line vertically

Without bias, the line would be forced to pass through the origin.

7. What is Loss?

Loss measures how wrong the model's predictions are.

It answers the question:

How far are predictions from the actual values?

The model's objective is to **minimize loss**.

8. Loss Function

A **loss function** is a mathematical formula that quantifies prediction error.

For each prediction:

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loss = error(prediction, actual)
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The training process tries to minimize the **average loss** across all samples.

9. Most Popular Loss Functions for Linear Regression

◆ Mean Squared Error (MSE) — most common

$$\text{MSE} = (1/n) \sum (\hat{y} - y)^2$$

Why MSE is popular:

- Penalizes large errors more strongly
 - Smooth and differentiable (good for optimization)
 - Widely used in practice
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◆ Mean Absolute Error (MAE)

$$\text{MAE} = (1/n) \sum |\hat{y} - y|$$

Characteristics:

- Less sensitive to outliers than MSE
 - Easier to interpret (average absolute error)
 - Not differentiable at zero (harder for optimization)
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◆ Root Mean Squared Error (RMSE)

$$\text{RMSE} = \sqrt{\text{MSE}}$$

Advantages:

- Same units as the label
 - Easier to interpret than MSE
 - Still penalizes large errors strongly
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◆ Huber Loss (less common, but useful)

Combines MSE and MAE:

- Acts like MSE for small errors
- Acts like MAE for large errors

Useful when the dataset contains outliers.

10. Training Objective

The goal of Linear Regression is to find the **optimal weights and bias** that minimize the chosen loss function.

This is typically done using:

- **Gradient Descent**
 - Or a **closed-form solution** (Normal Equation)
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11. Key Takeaways

- Linear Regression models a linear relationship between inputs and output
 - Features + weights + bias produce predictions
 - Loss measures prediction error
 - Training = minimizing loss
 - MSE is the most common loss for regression
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12. When to Use Linear Regression

Linear Regression works well when:

- Relationship between variables is approximately linear
- Data has limited noise
- Interpretability is important

It struggles when:

- Relationships are highly non-linear
- There are strong interactions between features